



— TECHNICAL DATA SHEET

NN-DustGuard PSPP

Polymer-based Surface & Stockpile Dust Suppressant



Supplied in 200 L HDPE drums and 1000 L IBC totes

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-01	1.0	27-06-2026	PSPP-01

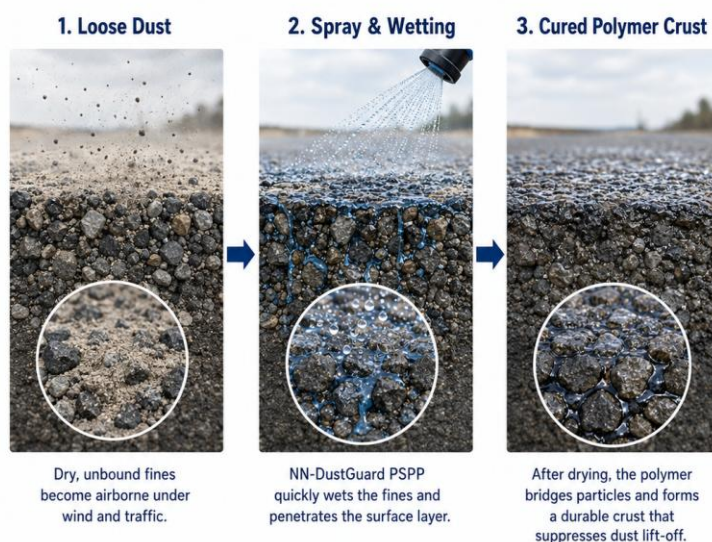
01 PRODUCT DESCRIPTION

NN-DustGuard PSPP is a water-based polymer emulsion concentrate engineered to control fugitive dust on open stockpiles, haul roads, unpaved surfaces, material transfer points and exposed earthworks. On application and curing it forms a flexible, weather-resistant crust that binds fine particles, resists wind erosion and rainfall washout, and substantially reduces airborne PM10 and PM2.5 emissions.

The product is supplied as a concentrate for dilution with water and is compatible with standard water-tanker spray bars, pressurised sprayers and dust-fog systems. It is non-corrosive to typical mild-steel and HDPE equipment and is formulated for the dust-control needs of cement, steel, power, mining, ports and large industrial sites.



HOW NN-DustGuard PSPP WORKS



Mechanism: fast wetting + particle penetration + polymer bridging + surface curing = durable dust suppression, reduced wind erosion, and improved road stability.



How NN-DustGuard PSPP works – wetting, penetration, polymer bridging and surface curing

02 KEY FEATURES & BENEFITS

- **Long-lasting crust** — Forms a durable, flexible polymer film that withstands wind and traffic, reducing re-application frequency versus plain-water spraying.
- **High dust capture** — Effective suppression of respirable and total dust, supporting compliance with ambient air-quality limits.
- **Weather resistant** — Resists wind erosion and rain washout; performs across a wide temperature and humidity range.
- **Water-dilutable** — Concentrate diluted on site (typical 1:5 to 1:20 with water) for flexible coverage and cost control.
- **Equipment friendly** — Non-flammable, water-based; compatible with existing tanker and spray infrastructure.
- **Environmentally considerate** — Free from added heavy metals and petroleum-oil binders; low-odour formulation.

03 SPECIFICATION & TYPICAL PROPERTIES

Property	Specification	Method / Unit
Appearance	Free-flowing liquid	Visual
Bio-degradable	Bio-degradable	—
Specific gravity	1.10 – 1.40 g/cc	Hydrometer
Density	1.10 g/cc and above	ASTM D1475



Property	Specification	Method / Unit
Solubility	100% soluble in normal water of pH 6.5 – 8.0 at NTP	—
Wetting / water holding	Minimum 70% water-holding capacity up to 24 hours of application at NTP	—
Agglomeration	Develops a minimum 2 mm agglomerated layer on the surface bed after three consecutive applications	Bed test
pH value	pH 6.5 – 8.0	IS 3025 / pH meter
Flash point	> 175°C	Closed cup
Air permeability	1.50 m ³ /m ² /min	—
Ionizing property	Present	—
Inflammable test	Non-inflammable	—
Shelf life	6 months from date of manufacture in unopened original packaging	—

04 APPLICATION & DOSAGE



Field application by NNPL tanker-mounted spray bar – bound vs. untreated surface

Parameter	Guideline	Notes
Dilution ratio	1:5 to 1:20 with water	Adjust to surface type & dust load
Coverage rate	1.0 – 2.5 L/m ² (diluted)	Higher for loose stockpiles
Application method	Spray bar / pressure sprayer / fog cannon	Apply evenly
Curing / set time	2 – 6 hours (climate dependent)	Avoid traffic until set
Re-application	Every few weeks (site dependent)	On visible crust breakdown

Directions: Loosen the surface where required, dilute the concentrate with clean water at the chosen ratio, and apply uniformly until the surface is wetted through. For stockpiles, apply to the point of light run-off without pooling. Allow to cure before resuming traffic.



05 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Packaging	50 L / 200 L HDPE drums; 1000 L IBC on request
Storage	Store in a cool, dry, shaded area between 5–40°C, away from frost and direct sunlight
Shelf life	12 months from date of manufacture in unopened original packaging
Handling	Keep containers closed; stir gently before use if settling occurs; do not allow to freeze

06 HEALTH, SAFETY & ENVIRONMENT

This product is a water-based emulsion classified as non-hazardous for transport under typical industry conditions. Wear gloves and eye protection during handling. Avoid discharge of undiluted concentrate to drains and watercourses. Refer to the accompanying Safety Data Sheet (SDS) for full first-aid, handling, spill and disposal guidance.

Disclaimer: The information in this TDS is given in good faith and is based on data believed to be reliable. NilayNarayan Polychem LLP makes no warranty, express or implied, as to its accuracy or completeness. Users must satisfy themselves as to the suitability and safe handling of the product for their own conditions and must comply with all applicable laws and regulations.